

1. Introduction

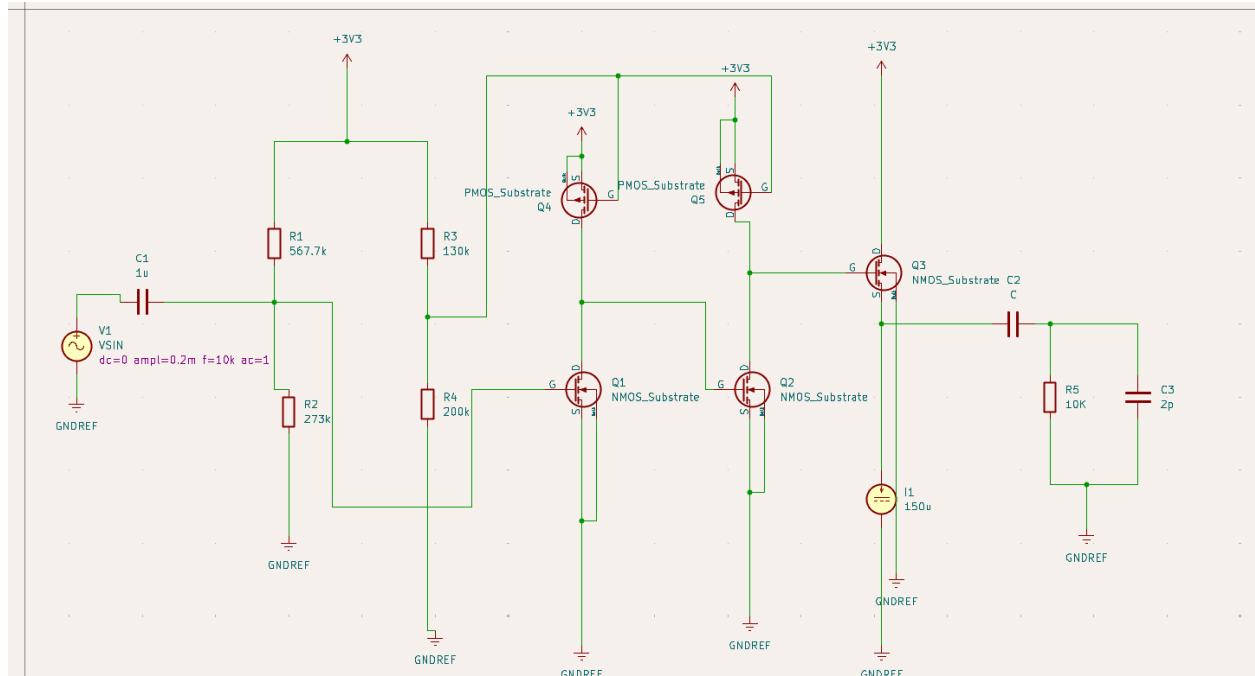
This project focused on designing an open-loop analog voltage amplifier using only MOSFETs and verifying its performance in KiCad. The amplifier needed to meet specific requirements for gain, bandwidth, output swing, input resistance, and power. Since these requirements interact with each other, the design process aimed to find a practical balance instead of just maximizing one parameter. We used manual calculations as a starting point and then employed KiCad simulations to confirm and refine the final circuit.

2. Design Architecture

The final circuit features a three-stage CS-CS-CD structure, which consists of two common-source stages followed by one common-drain stage. We chose this topology because a single stage would not provide enough gain, while the final stage was necessary to buffer the output and improve load driving.

The first two stages are common-source amplifiers that use NMOS transistors as amplifying devices and PMOS transistors as active loads. These stages produce most of the voltage gain. The third stage is a common-drain stage, also known as a source follower, that is biased with a current source. Its main role is to buffer the output rather than add significant voltage gain.

The circuit also includes input and output coupling capacitors. The input capacitor blocks unwanted DC from the source, while the output capacitor prevents the amplifier bias from being transferred to the load. Internally, the stages are directly connected, allowing the DC operating point of one stage to bias the next.



Figure#1: CS-CS-CD MOSFET amplifier schematic designed for the project

3. Experimental Results/Simulations

Table 1: Performance Results for Theoretical vs. KiCad Simulation

	Parameters	Constraints	Theoretical Results	Experimental Results	Pass? (Y/N)
Input	Supply Voltage	3.3V	3.3V	3.3V	Y
	Signal Amplitude	$\leq 10 \text{ mVpp}$	—	0.2 mVpp	Y
	Input Resistance	$\geq 100\text{k}\Omega$	195.7 k Ω	184.35 k Ω	Y
Gain	Total open-loop Gain (unloaded)	$\geq 60\text{dB}$ ($\geq 1000\text{V/V}$)	80dB	65.7dB	Y
	Total open-loop Gain (loaded)	—	—	65.1dB	—
	Gain Reduction	$\leq 10\%$	9.6%	6.67%	Y
	Gain per stage	$\leq 40\text{dB}$	Stages 1 & 2: 40dB Stage 3: 0dB	Stage 1: 36dB Stage 2: 31dB Stage 3: -1.9dB	Y
Bandwidth	-3 dB Bandwidth	$\geq 500\text{kHz}$	27 MHz	1.25 MHz	Y
Output	Output Swing	$\geq 1.5\text{Vpp}$	—	1.53Vpp	Y
Power	Total DC Power	$\leq 1\text{mW}$	0.86mW	0.92mW	Y
	Max branch current	$\leq 200\mu\text{A}$	150 μA	150 μA	Y
	Transistor Length	$0.5\mu\text{m} \leq L \leq 5\mu\text{m}$	0.5 μm	0.5 μm	Y
	Overdrive Voltage	$0.2\text{V} \leq V_{ov} \leq 0.6\text{V}$	$V_{ovn} = 0.25\text{V}$ $V_{ovp} = 0.30\text{V}$	$V_{ovn} = 0.44\text{V}$ $V_{ovp} = 0.57\text{V}$	Y

Below are the relevant simulation plots that verify gain, bandwidth and output swing:

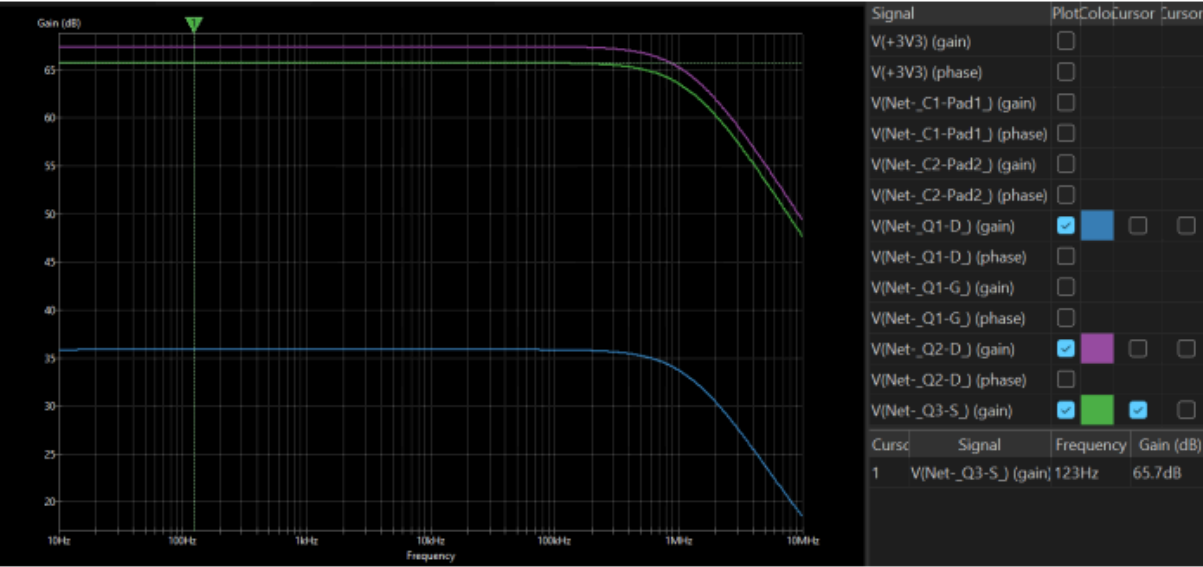


Figure 2: AC analysis for unloaded gain ($R_{out} = \infty$)

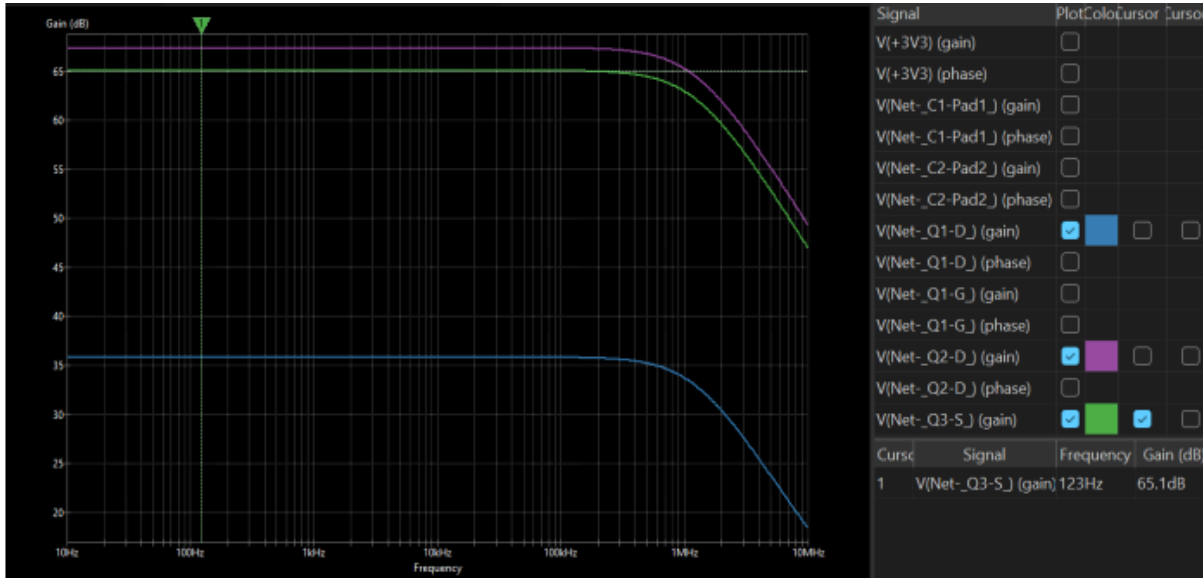


Figure 3: AC analysis for loaded gain ($R = 10k\Omega$)

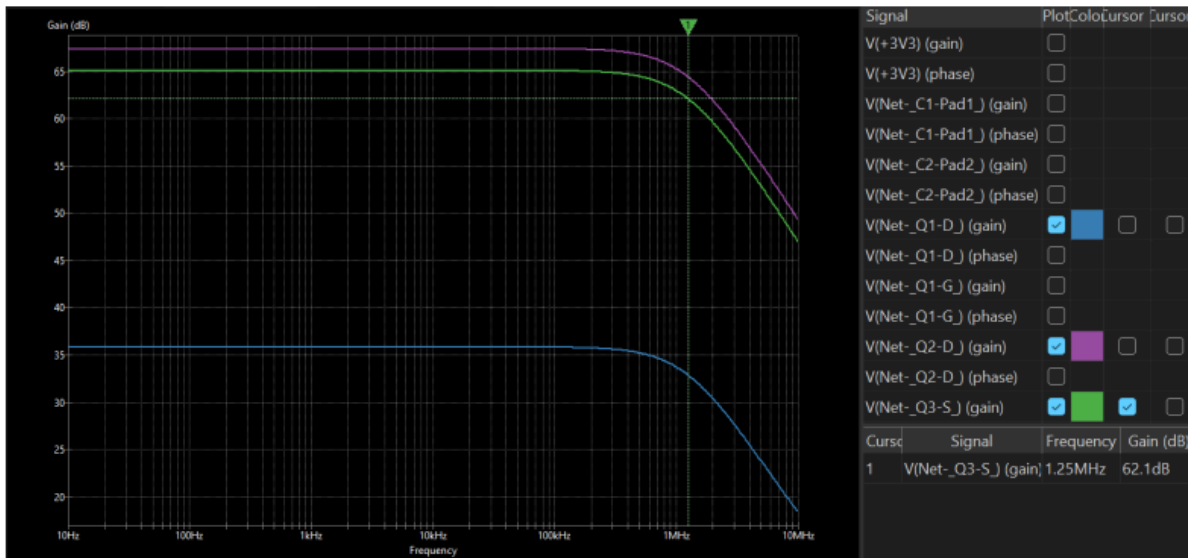


Figure 4: AC analysis for loaded -3dB bandwidth

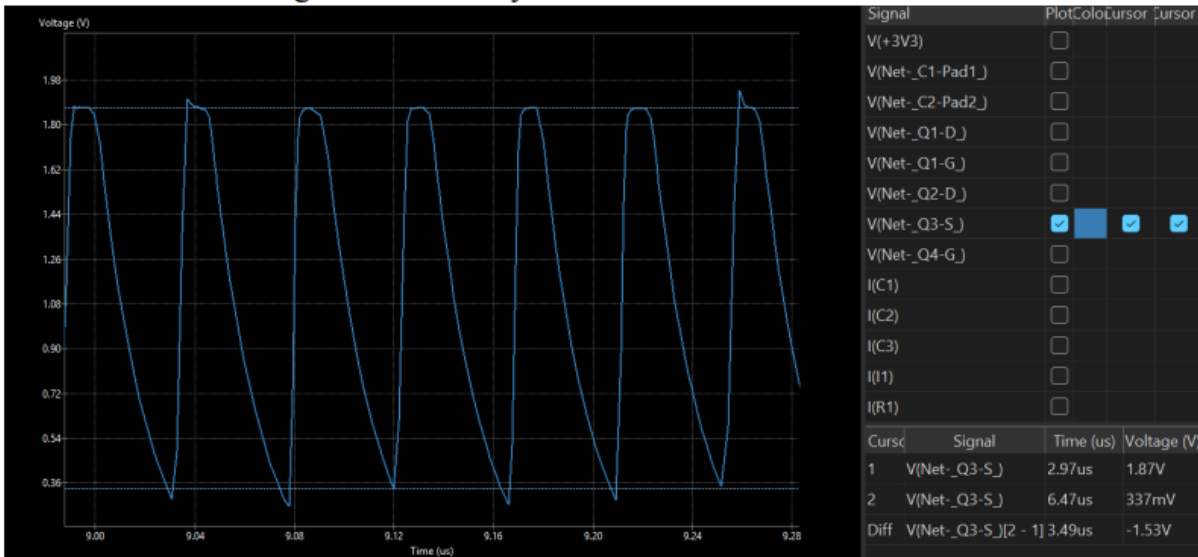


Figure 5: Transient analysis for output swing

After designing the circuit, I simulated it in KiCad and compared the results with theoretical expectations. The final amplifier met all major requirements.

The simulated input resistance was 184.35×10^3 ohms, which exceeds the required minimum.

The open-loop gain was 65.7 dB unloaded and 65.1 dB loaded, meeting the required gain. The gain reduction due to loading was 6.67 percent, which is within the acceptable limit. The stage

gains aligned with the intended operation; the first two stages contributed most of the gain, while the third stage mainly operated as a buffer.

The simulated -3 dB bandwidth was 1.25×10^6 Hz, surpassing the required minimum. The output swing was 1.53 Vpp, and total DC power was 0.92×10^{-3} W, remaining within the power limit. The maximum branch current was 150×10^{-6} A, also within the allowed range. Overall, the simulation confirmed that the amplifier satisfied the design constraints.

4. Analysis / Discussion

The simulated results were reasonably close to the theoretical calculations, but they were not exactly the same. This was expected because the manual calculations used simplified MOSFET equations, while the KiCad simulation applied more detailed transistor models and accounted for non-ideal effects. As a result, the hand calculations were mainly useful for selecting a good starting point, while the simulation offered a more realistic view of the circuit's actual performance.

One noticeable difference was in the overall gain. The theoretical design predicted a higher gain, but the simulated gain was lower because the real circuit included stage loading and parasitic capacitances. In a multistage amplifier, each stage influences the next one, so the stages do not behave as ideally as they do in manual calculations.

Another important point was the tradeoff between gain and bandwidth. The first two common-source stages were necessary to achieve the required gain, but increasing the gain also made the internal high-resistance nodes more sensitive to capacitance, which reduced bandwidth. This meant the transistor sizes had to be chosen carefully so that both the gain and bandwidth requirements could be satisfied.

The output stage was also crucial to the final design. Even though the common-drain stage did not provide much voltage gain, it served as a buffer and minimized the loading effect of the output network on the earlier stages. Overall, the final CS-CS-CD architecture struck a good balance between gain, bandwidth, output buffering, and power.

5. Conclusion

In summary, the final amplifier fulfilled the ELE404 project requirements using a three-stage CS-CS-CD MOSFET design. The first two stages delivered the needed voltage gain, while the final common-drain stage served as a buffer to improve output driving capacity. KiCad simulation confirmed that the circuit met the necessary limits for gain, bandwidth, input resistance, output swing, power, and current. Although the simulated values did not perfectly match the theoretical calculations, the overall results were reasonable and demonstrated the effectiveness of the design process.

6. Appendix

Theoretical Calculations:

$$\text{Max Current: } I_{\max} = \frac{1\text{mW}}{3.3\text{V}} = 303\mu\text{A}$$

Current distribution and power:

$$\text{Stage 1 (Common Source): } I_{D1} = 50\mu\text{A}$$

$$\text{" 2 (" ") : } I_{D2} = 50\mu\text{A}$$

$$\text{Stage 3 (Common Drain): } I_{D3} = 150\mu\text{A}$$

$$\text{Bias Current: } I_{\text{Bias}} = 10\mu\text{A}$$

$$\text{Total Current: } I_{\text{total}} = 260\mu\text{A} \text{ or } P_{\text{tot}} = 0.86\text{mW} \leq 1\text{mW}$$

Stage 1 & 2 Common Source Amps

Overdrive Voltage Chosen ($0.2 \leq V_{ov} \leq 0.6$):

$$V_{ovn} = 0.25\text{V} ; V_{ovp} = 0.3\text{V}$$

Transistor Sizes ($L = 0.5$ for all transistors):

$$I_D = \frac{1}{2} \mu C_{ox} \left(\frac{W}{L}\right) V_{ov}^2 \quad ; \quad \mu_n C_{ox} = 100\mu\text{A/V}^2 \quad ; \quad \mu_p C_{ox} = 25\mu\text{A/V}^2$$

NMOS:

$$50\mu\text{A} = \frac{1}{2} (100\mu\text{A/V}^2) \left(\frac{W_n}{0.5}\right) (0.25\text{V})^2$$

$$\text{Isolate for } W_n: W_n = 8\mu\text{m}$$

PMOS:

$$50\mu\text{A} = \frac{1}{2} (25\mu\text{A/V}^2) \left(\frac{W_p}{0.5}\right) (0.3\text{V})^2$$

$$\text{Isolate for } W_p: W_p = 22.2\mu\text{m}$$

Gain:

$$g_{m1} = g_{m2} = \frac{2I_D}{V_{ov}} = \frac{2(50)}{0.25} = 400\mu\text{S}$$

$$r_{on} = r_{op} = \frac{1}{\lambda I_D} = \frac{1}{0.04(50)} = 500\text{k}\Omega$$

$$R_{out1} = R_{out2} = r_{on} \parallel r_{op} = 250\text{k}\Omega$$

$$A_{v1} = A_{v2} = -g_m \cdot R_{out} = -(400)(250) = -100\text{V/V}$$

$$A_{0dB} = 20 \log_{10}(100) = 40\text{dB} \leq 40\text{dB}$$

DC Bias (Voltage Divider):

Input resistance and NMOS Gate Bias:

$$V_{G1} = V_{tn} + V_{ovn} = 0.63V + 0.25V = 0.88V \text{ (required gate bias)}$$

Since $R_{in} \geq 100k\Omega$, assume $R_1 + R_2 = 1M\Omega$

$$V_{G1} = V_{DD} \left(\frac{R_2}{R_1 + R_2} \right) \Rightarrow 0.88 = 3.3 \left(\frac{R_2}{1M\Omega} \right) \rightarrow R_2 = 267k\Omega, R_1 = 733k\Omega$$

$$R_{in} = R_1 \parallel R_2 = 195.7k\Omega \geq 100k\Omega$$

PMOS Gate Bias:

$$V_{SG} = |V_{tp}| + V_{ovp} = 0.73V + 0.3V = 1.03V \text{ (required source gate bias)}$$

$$V_G = 3.3V - 1.03V = 2.27V$$

Bias Branch / voltage divider (Chose to allocate $10\mu A$ current):

$$R_{bias} = \frac{3.3V}{10\mu A} = 330k\Omega$$

$$R_4 = 330 \frac{2.27}{3.3} = 227k\Omega ; R_3 = 330 - 227 = 103k\Omega$$

Stage 3: Common Drain Amp.

Chosen current and V_{ov} : $I_{D3} = 150 \mu A$; $V_{ov} = 0.3 V$

NMOS:

$$150 \mu A = \frac{1}{2} (100 \mu A/V^2) \left(\frac{W_n}{0.5} \right) (0.3 V)^2 \rightarrow W_{n3} = 16.7 \mu m$$

$$g_{m3} = \frac{2(150 \mu A)}{0.3 V} = 1000 \mu S = 1 mA/V$$

Verify load sensitivity

$$R_L = 10 K \Omega$$

$$r_{o3} = \frac{1}{0.04 (150 \mu A)} = 166.7 K \Omega$$

$$A_{v3} (\text{loaded gain}) = \frac{R_L \parallel r_{o3}}{(R_L \parallel r_{o3}) + \frac{1}{g_{m3}}} = \frac{9.4 K \Omega}{9.4 K \Omega + 1 K \Omega} = 0.904 V/V$$

$$\text{Gain reduction: } (1 - 0.904) \cdot 100 = 9.6\% \leq 10\%$$

Total Gain:

$$A_{v \text{ total}} = A_{v1} \cdot A_{v2} \cdot A_{v3} = (-100)(-100)(1) = 10,000 V/V$$

$$A_{dB, \text{ total}} = 20 \log(10000) = 80 dB \geq 60 dB$$

Bandwidth Calculation:

$$C_{ox} = 4.3 fF/\mu m^2 \quad C_{ov} = 0.4 fF/\mu m$$

$$C_{GS2} = \frac{2}{3} C_{ox} W_n L = \frac{2}{3} (4.3) (8) (0.5) = 11.5 fF$$

$$C_{GD1} = C_{ov} W_n - (0.4) (8) = 3.2 fF$$

$$C_{GD2} = C_{ov} W_n - (0.4) (22.2) = 8.9 fF$$

$$C_{\text{node}} = 11.5 + 3.2 + 8.9 = 23.6 fF$$

$$f = \frac{1}{2\pi (R_{\text{node}}) (C_{\text{node}})} = \frac{1}{2\pi (250 K) (23.6 fF)} = 27 MHz$$